



Agragaman - 2019

(The Precession)



Centre for Advanced Studies
(Dr. A. P. J. Abdul Kalam Technical University, Uttar Pradesh, Lucknow)

2nd Raising Day Celebration
7th July 2019

Centre for Advanced Studies

(Dr. A. P. J. Abdul Kalam Technical University
Uttar Pradesh, Lucknow)

कुलगीत

हमारा विश्वविद्यालय, प्रगति सोपान का उद्गम।
प्राविधिक उच्च शिक्षालय, विवेक और ज्ञान का संगम॥
हमारा विश्वविद्यालय

विश्वव्यापी धरातल पर, मनुज-उत्थान का सपना।
नियोजन, श्रमजी, अभियांत्रिकी, वरदान हो अपना।
कुशल प्रबंधन, सही सोच को, हम सब करें हृदयंगम॥
हमारा विश्वविद्यालय

सृजन की तीव्र उत्कंठा, सभी मन में समाई हो।
लगे परिवार-सम हम सब, प्रगति की लहर छाई हो।
बने 'सम्पूर्ण गुणवत्ता', हृदय की कामना अनुपम॥
हमारा विश्वविद्यालय

हमारी प्राविधिक शिक्षा, सुमानव-मूल्य रखती हो।
हमारा ध्येय, महिमा देश की, दुनिया निरखती हो।
अमर गुण-ज्ञान गरिमा से, मिटे अज्ञानता का तम।
हमारा विश्वविद्यालय

नई चेतना से प्रेरित हम, लगे जीव-कल्याण में।
मानवता हित तकनीकी से, जुटे राष्ट्र-निर्माण में।
बने प्रेरणा-स्रोत हमारा; योगः कर्मसु कौशलम्॥
हमारा विश्वविद्यालय





आशुतोष टण्डन "गोपाल जी"

मंत्री

प्राविधिक शिक्षा एवं
चिकित्सा शिक्षा विभाग



कार्यालय : 2238896

सी.एच. : 2213251

कक्ष संख्या 59-59ए, मुख्य भवन
विधान भवन, लखनऊ



दिनांक 05/07/19



संदेश

मुझे यह जानकारी प्रसन्नता हुई है कि डा० ए०पी०जे० अब्दुल कलाम प्राविधिक विश्वविद्यालय उ०प्र०, लखनऊ का घटक संस्थान सेंटर फॉर एडवांस्ड स्टडीज, लखनऊ दिनांक 07 जुलाई, 2019 को अपना द्वितीय स्थापना दिवस समारोह मना रहा है। वर्ष का विषय है कि संस्थान कम्प्यूटर साइंस एण्ड इंजीनियरिंग विद स्पेशलाइजेशन इन साइबर सिक्योरिटी, मेकाट्रानिक्स, नैनोटेक्नोलॉजी, मैनुफैक्चरिंग टेक्नोलॉजी एण्ड आटोमेशन तथा एनर्जी साइंस एण्ड टेक्नोलॉजी जैसी प्रासंगिक शाखाओं में परास्नातक पाठ्यक्रम संचालित कर रहा है।

संस्थान के द्वितीय स्थापना दिवस समारोह के सुअवसर पर मेरी हार्दिक शुभकामनाएं।

(आशुतोष टण्डन)



भुवनेश कुमार
सचिव



अ0शा0प0सं0- /प्रा0शि0वि0/2019
उत्तर प्रदेश शासन
प्राविधिक शिक्षा विभाग
दूरभाष नं0: 0522-2238094 (का0)
लखनऊ

दिनांक ०५-०७-२०१९



संदेश

मुझे यह जानकर अत्यन्त हर्ष हुआ है कि समाज के भौतिक और आध्यात्मिक ताने-बाने को व्यवस्थित करने का कार्य उच्च शिक्षा का है। कौशल एक कला है जो व्यक्ति को सृजनात्मकता का क्रियात्मक आचरण 'तकनीक' का निर्माण करता है। अतः तकनीकी शिक्षा का मुख्य ध्येय ऐसे मानव संसाधन का निर्माण करना होना चाहिए जो अपनी सृजनात्मकता के क्रियात्मक आचरण से समाज के सकारात्मक विकास के लिए नित उन्नत तकनीक का निर्माण कर सके। डॉ० ए०पी०जे० अब्दुल कलाम प्राविधिक विश्वविद्यालय उत्तर प्रदेश, लखनऊ के घटक संस्थान सेंटर फॉर एडवांस स्टडीज, लखनऊ द्वारा कम्प्यूटर साइंस एंड इंजीनियरिंग की प्रासंगिक ब्रांचों जैसे सिक्योरिटी, नेनो टेक्नोलॉजी एवं मैकेट्रानिक्स परास्नातक स्तर की विश्वस्तरीय गुणवत्तापरक शिक्षा प्रदान करा रहा है। हर्ष का विषय है कि संस्थान 7 जुलाई, 2019 को अपना द्वितीय स्थापना दिवस मना रहा है। द्वितीय स्थापना दिवस के सुअवसर पर मैं, विश्वविद्यालय के कुलपति, संस्थान के निदेशक, शिक्षकों, अधिकारियों, कर्मचारियों एवं छात्र-छात्राओं को शुभकामनाएं देता हूँ और संस्थान एवं उसके छात्र-छात्राओं के उज्ज्वल भविष्य की कामना करता हूँ।

भुवदीय

(भुवनेश कुमार)

श्री प्रो० मनीष गौड़,
निदेशक,
सेंटर फार एडवांस स्टडीज,
डॉ० ए०पी०जे० अब्दुल कलाम प्राविधिक विश्वविद्यालय,
उत्तर प्रदेश, लखनऊ।



प्रो० विनय कुमार पाठक
कुलपति
Prof. Vinay Kumar Pathak
Vice-Chancellor



डॉ० ए०पी०जे० अब्दुल कलाम प्राविधिक विश्वविद्यालय
उत्तर प्रदेश, लखनऊ
Dr. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY
Uttar Pradesh, Lucknow



03.07.2019

संदेश

अत्यंत हर्ष का विषय है कि सेंटर फार एडवांस्ड स्टडीज, लखनऊ अपना द्वितीय स्थापना दिवस 07 जुलाई, 2019 को मनाने जा रहा है और इस अवसर पर अपनी विगत एक वर्ष की प्रगति से संबंधित विवरणिका अग्रगमन को भी प्रकाशित करने जा रहा है। विवरणिका का प्रकाशन निश्चय ही एक अच्छा प्रयास है।

मैं आशा करता हूँ कि यह कार्यक्रम सफलतापूर्वक सम्पन्न हो एवं सेंटर फार एडवांस्ड स्टडीज, लखनऊ उत्तरोत्तर प्रगति के पथ पर अग्रसर होते हुए अपने निर्धारित लक्ष्य को प्राप्त करे। मुझे विश्वास है कि यह संस्थान प्रदेश में तकनीकी शिक्षा के गुणवत्ता सुधार एवं प्रदेशोन्मुखी शोध कार्यों में अपना महत्वपूर्ण योगदान देगा। आप सभी को पत्रिका के सफल प्रकाशन हेतु शुभकामनाएं।

Vinay Kumar Pathak

(प्रो० विनय कुमार पाठक)
कुलपति



CENTRE FOR ADVANCED STUDIES

Dr. APJ Abdul Kalam Technical University Uttar Pradesh, Lucknow

Prof Manish Gaur

Director, CAS
Fellow (University of Glasgow, UK),
Ph.D. (University of Sussex, UK),
M.Tech. (IIT Delhi, India), B.E. (NIT Surat, India)



05.07.2019

Message

It is indeed a matter of great pride and pleasure that Centre for Advanced Studies at AKTU is completing two years of its establishment. I am one of the few lucky ones who have witnessed the growth of this Institution since its inception. In a very small period the Institute has grown leaps and bounds; it all started with 14 students registering for M.Tech programme in Computer Science & Engineering with Cyber security specialization. Now the current strength of students has risen up to 54. I am extremely glad to share that two new masters programs, both in the areas of national importance, namely Manufacturing Technology & Automation and Energy Science & Technology are all set to start from the forthcoming academic session. The institute never compromises in terms of quality of teaching research and other academic endeavours. Such journeys are never easy but with the blessings and support of Hon'ble Vice Chancellor we could surpass all the obstacles. The institute is making its mark towards research, development and quality improvement of teaching through its world class infrastructures and manpower. The institute has to take a long journey and the first step towards it has already been successfully taken.

We, the faculty members, research engineers, staff and students are thrilled on the completion of two years of the Institute. On this occasion a newly built Materials Chemistry and Synthesis lab will be commissioned. This shall be a great augmentation in our state of art laboratory facility. I look forward for the completion of ongoing student's hostel and faculty residences as soon as possible to make this institution residential.

The launch of the yearly information booklet namely "AGRAGAMAN-2019" is a useful compilation of all the achievements and future goals.

I wish a grand success of the establishment day celebration.

(Prof. Manish Gaur)

Centre for Advanced Studies

Dr. APJ Abdul Kalam Technical University U.P., Sector 11, Jankipuram Extension, Lucknow, UP 226031

Website: www.aktu.ac.in/cas/, Email-id: director.cas@aktu.ac.in, Phone No.-0522-2771079, Mob No: 9415039258



About the Institute

The Technical University system has to be the main torch bearer of engineering education and research in the state. It is necessary to invest at the appropriate level for cultivating, on a sustainable basis, the ethos of engineering education and research. Uttar Pradesh Government issued a notification to start Centre for Advanced studies for upgrading the standards of academics, quality of staff and research in the state. To make Dr. APJ Abdul Kalam Technical University as an institution of higher learning of international repute, a Centre for Advanced Studies (CAS-AKTU) as in-campus academic & research institution was setup in 2017 and the Institute has started functioning from July 2017. The institute in its first phase started one M.Tech. programs in Computer Science & Engineering with specialised areas of Cyber Security and Information and Communication Technologies (ICT). Later in 2018, the Institute has started two other M.Tech. programs in Mechatronics and Nano-Technology. From session 2019-20, the institute has started two more programs in Manufacturing Technology & Automation and Energy Science & Technology. Apart from PG Programs, the institutes also has PhD. Programs in Computer Science & Engineering, Mechatronics and Nano-Technology. The Institute envisages to start courses and research in the frontier areas such as Bio Technology, 3D printing, Climate control, Market and Business Analytics, Big Data, to name a few.

Our Vision

Centre for Advanced Studies will create a better future for all through its leadership in research and excellence in engineering education. It will accomplish this by fostering a challenging, enlightened, and collaborative academic environment that demands excellence, encourages innovative education, and supports ground-breaking discovery. Its esteemed faculty will be world renowned for their creative scholarship and innovative teaching. The institute's graduates will be recognized and valued for their commitment to excellence, enthusiasm for learning, ethical behaviour and integrity, and exceptional leadership.

Our Mission

- To establish Centre for Advanced Studies of Dr. APJ Abdul Kalam Technical University as a world class centre of Excellence for Postgraduate Studies as well as cutting edge Research and Development in the areas of Engineering & Technology.



- To provide a stimulating platform to research scholars and academicians in Engineering & Technology domain; for creating and disseminating research-based knowledge/technologies for the development of State/Country.
- To bring together creative, dedicated & innovative scholars engaged in Engineering and Technology academics and research.

Aims & Objectives

The aim of the “Centre for Advanced Studies” will be to impart state of the art education leading to Post Graduate Degree as well as to facilitate quality research work in the emerging areas of the technology to establish Dr. APJ Abdul Kalam Technical University as a world class premier academic & research institution in the state with a global perspective.

- To create a unique institution in the country where in engineering & technology education will be totally integrated with state-of- the-art research
- To create, therefore, an Integrated postgraduate degree programme in engineering following graduation
- To create a cadre of high-calibre, internationally well-known faculty members which will be devoted to teaching as well as research activities in engineering & Technology
- To strive to arrest the declining trend of young people in joining engineering education & research
- To elevate the status of the university in terms of education & research in engineering & technology, at par with IITs & IIMs
- To provide flexibility, education and training in order to charter new grounds and break compartmentalization of education & research under the domain of Technology & Engineering
- To make possible a flexible, border-less curriculum in which a student, say interested primarily in biology, should have no difficulty in taking courses of other subject domains
- To Collaborate or sign up MOU with International as well as Nation universities and institutions of repute.



- To network Centre for Advanced Studies (CAS-Dr. APJ Abdul Kalam Technical University) with the existing institutions, laboratories, universities and colleges in the neighbourhood in order to make optimal utilization of resources, both in terms of expensive laboratory facilities as well as teaching talents
- To build a strong core faculty (permanent, Guest, Emeritus) that will be able to synergize research with teaching and education
- To develop world-class research laboratories in order to improve the quality of education as well as research in Uttar Pradesh.

Highlights of Centre for Advanced Studies

- Fully autonomous for all academic endeavours.
- Highly Qualified and Experienced faculty.
- Research oriented curriculum and high-tech state of the art Labs.
- Opportunity for Internship/ Dissertation in National and International organizations of repute.
- Exposure to International faculty / Scientist of repute.
- High Tech Campus with all modern facilities with speed Wi-Fi.
- Full Campus with high speed Wi-Fi connectivity.
- Centrally Air-conditioned campus.
- Google Sponsored Google Code Developers Lab.
- Cyber Physical Simulator lab, first of its kind in North India.
- Digital Library with access to International Science and Engineering books and Journals including E-resources.
- KUKA Robotic Centre (Installation under process)
- Industrial Automation Lab (Installation under process)
- 3-D printing Lab (Installation under process)
- Many advanced research labs are under process of installation.
- Focus on extra-curricular activities for holistic development of the students.



- Methods of teaching includes-learning by doing, workshop, and seminars.
- Credit options on MOOCs.
- All current M.Tech students are doing their summer internship in institute/industry of repute like IIScBagalore, IIT Kanpur, IIT Patna, IIT Guwahati, KUKA robotic Centre Ghaziabad etc.

Courses Offered

M.Tech. in

- Computer Science & Engineering with specialization in
 - Cyber Security (started from 2017-18 session)
 - Information and Communication Technologies (started from 2017-18 session)
- Mechatronics (started from 2018-19 session)
- Nanotechnology (started from 2018-19 session)
- Manufacturing Technology & Automation (To be started from 2019-20 session)
- Energy Science & Technology (To be started from 2019-20 session)

Ph.D course in

- Computer Science & Engineering
- Mechatronics
- Nanotechnology

Residential Facility

Boys and Girls Hostel of 75 rooms each and 32 faculty & staff quarters (Type III & IV) are under construction. After the construction the campus would become fully residential.





Prof. Vinay k Pathak

Vice Chancellor

Dr. A.P.J. Abdul Kalam Technical University
Lucknow

Ph.D. (UPTU Lucknow)
M.Tech. (IIT Kharagpur)
B.Tech. (HBTI Kanpur)

Area of Interest:

Image Processing, Computer vision,
Pattern recognition



Prof. Manish Gaur

Director

Center for Advanced Studies

Research Fellow of
University of Glasgow, UK
Ph.D., University of
Sussex, UK
M.Tech. (IIT Delhi)
B.Tech. (NIT Surat)

Area of Interest:

Formal Methods and Verification of Large
Systems in process algebraic framework,
Semantics of programming Languages in the
domain of Computer Science.



Dr. M. K. Dutta
Professor
Dean R&D
and Industrial
Consultancy,
AKTU

Ph.D.
(Gautam Buddha
Technical
University)
M.Tech
(Gold Medallist)
Central University
Tezpur.

Area of Interest:

- Image Processing, Computer Vision & Pattern Recognition.
- Medical Informatics.
- Audio Signal processing.
- Bio-Medical Signal Processing applications.
- Multimedia Signal Processing : Tools and Applications
- Data Security: Digital Watermarking & Encryption
- Electronic Security and Digital Forensics.
- Applications of Signal Processing in Food Science, Nano-Science and Bio-Technology.



Dr. Piyush Jaiswal
Coordinator
Nanotechnology

Ph.D.,
Indian Institute of
Science(IISc),
Bangalore.
M. Tech,
NIT Allahabad.
B.Tech,
I.E.T.,
Rohilkhand
University, Bareilly

Area of Interest:

MOCVD and ALD of oxide thin films, Microwave irradiation assisted deposition of nanostructures and thin film Nano-composite hard coating for machining, Microstructure development and properties of CVD thin films, development of precursors for (MO)CVD process, Micro- and nano-structure development using pyrolysis method(s).



Dr. Anuj K. Sharma
Coordinator
Mechatronics
and Associate Dean
R&D and Industrial
Consultancy, AKTU

Ph.D., IIT (ISM)
Dhanbad.
M. Tech., U.P.T.U,
Lucknow.
B.E., Chaudhary
Charan Singh
University,
Meerut, UP.

Research Interest:

Metal cutting operations
Electrical discharge machining
Machining tribology
MQL Machining
Nano-cutting fluids
Thermocouples for machining
Cutting speed controller for turning operation.



Dr. Ramkrishna Sahoo
DST Inspire
Faculty

Post-doctoral Fellow,
Sungkyunkwan
University Korea
PhD IIT Kharagpur
MSc West Bengal
State University
BSc. University
of Calcutta

Area of Interest:

Use of Nano science and nanotechnology to fabricate advanced electro-chemical energy storage and conversion devices



Dr. Manish Raj
Assistant Professor

Ph.D. IIIT,
Allahabad).
M.Tech. IIIT
Allahabad).
B.Tech. (UPTU,
Lucknow).

Area of Interest:

Robotics and Automation,
Control System, Non-linear
Dynamics, Machine learning,
Optimization, Mechatronics, Digital
Signal Processing, Humanoid
Robotics

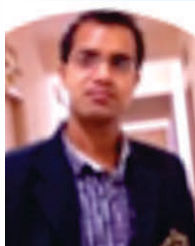


Dr. Rabesh Kumar Singh
Assistant Professor

PhD IIT (ISM),
Dhanbad.
M. Tech IIT (ISM),
Dhanbad.
B.Tech
GLA Institute of
Technology and
Management,
Mathura

Area of Interest:

Metal cutting operations Die-sink
EDM Tribology of machining.
Machining with MQL Nano-cutting
fluids Sensors for measurement of
Temperature and Cutting forces in
metal removal processes



Dr. Saurabh Mishra
Assistant Professor

B.Tech. Uttar
Pradesh Technical
University, Lucknow.
M.Tech & Doctorate
from IIT (ISM),
Dhanbad.

Area of Interest:

Sand Control Methods at oil fields,
Enhanced Oil Recovery,
Application of nanotechnology
in the oil industry.



Dr. Sajal Agarwal
Assistant Professor

Ph. D MNNIT,
Allahabad
M.Tech.
VNIT, Surat.
B.Tech. SRMSCET,
Bareilly

Area of Interest:

Thermally evaporated thin films,
surface plasmon resonance-based
biosensor, application of nano-
materials in solar cells, thermo-
photovoltaic cells, optical
communication, optical grating-
based sensors.



**Dr. Neelam
Dayal**
Assistant Professor

Ph.D. (Computer
Science &
Engineering),
MNNIT Allahabad.
M.Tech. (Mobile
Computing),
NIT Hamirpur

Area of Interest:

Software Defined Networking,
Network Security, Cryptography,
Wireless Sensor Networks, Mobile
Computing.



Dr. Amrit Pal
Assistant Professor

Ph.D. IIIT, Allahabad).
M.Tech. (Computer
Technology and Application)
(NITTTR, Bhopal).
B.Tech. (Kurukshetra
University, Kurukshetra)

Area of Interest:

Machine Learning and
Data Analytics, Big Data
Analytics, Cloud Computing
Internet of Things.



**Dr. Vrijendra
Singh**
Visiting Faculty

Ph.D., Dayalbagh
Educational
Institute, Agra.
M.Sc., in 2002,
Dayalbagh
Educational
Institute, Agra

Area of Interest:

Data Mining, Machine Learning,
Optimization Techniques, Modelling &
Simulations. Blind Source Separation,
Digital Signal and Image Processing,
Information Security & Forensics,
Materials Management & Inventory
Control, Intelligent Enterprise
Resource Planning, Project Management.



Staff Members



**Mr. Veer
Vikram**
Research
Engineer

**BSc. (CS)
MSc (IT)
MCA**

Area of Interest:
Computer Networking



**Mr. Diyanshu
Chauhan**
Research Engineer

Masters from Guru
Gobind Singh
Indraprastha University
New Delhi in
Computer application
and industrial training
from CDAC.

Area of Interest:
Web Applications Development,
Database Systems, AI,
Software Testing,
Operating Systems.



Mr. Gaurav Rai
Research
Engineer

**M. Tech.,
Amity University
PG. Diploma,
CDAC Acts, Pune.**

Area of Interest:
Embedded System, Real Time
Operating System. Design Micro
control and Microprocessor Based
systems, FPGA and ASIC design,
Frontend and Backend IC
designing. Sensor based system,
Internet of Things, VHDL and
Verilog.



Dr. Ram Kumar Pathak
Library
in- Charge
Central Library

B. Sc with
(Physics honors);
M.Sc. (PHY with
specialization in
solid state electronics)
BLIS MLIS; PGDLAN
UGC-NET

Ph.D. from Assam
University (Central
University), Silchar,
Assam.

Area of Interest:

E-Learning, e-Library, Digital library, knowledge management, library automation, library policy and management, personnel management, informatics, non-numerical multimedia database management; Designing and implementing library systems including technology and organizational process in addition to building strong teams.



Sadka Kouser
Programmer

M.C.A.

Area of Interest :

Database server, web applications, JQuery, web api integration and development, wcf, bootstrap, cryptography, system security, Angular.



Sadka Kouser
Programmer

B. Tech.

Area of Interest :

Database server, web applications, JQuery, web api, bootstrap, MVC, JSON, Angular.

Administration Staff

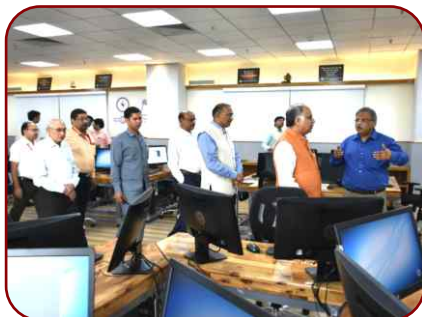
Mr. Sunil K Pandey	Assistant Registrar
Mr. Mukesh Kumar Mourya	PA to Director



State of the Art Infrastructure

➤ **GOOGLE DEVELOPERS CODE LAB**

GOOGLE has established a lab in Centre for Advanced Studies where students can play around with their crazy ideas and get feedback directly from Google's developers. The lab is equipped with high configuration system with dual monitor screens for better coding experience. The lab is a state of art facility in Uttar Pradesh and provide a full environment of working in an organization. This lab has a student capacity of 40 and also suitable for training programs. The lab is currently used by the students CAS-AKTU and other affiliated institutes of AKTU in state of UP for short term training/internship programs.



➤ **FIBRE OPTIC COMMUNICATION NETWORK LAB**

➤ **OPEN SOURCE LAB**

➤ **LANGUAGE LAB**

➤ **CYBER SECURITY LAB** (Cyber physical system simulator for security testing & training) Cyber City Simulator lab dedicated to train the students to discover vulnerabilities on smart devices in a simulated cyber city.



➤ **HIGH PERFORMANCE COMPUTING LAB**

➤ **CENTRAL LIBRARY**

- State of the art Central library RFID Enabled circulation having e-books accessibility of 24x7.





- Library contains more than 7000+ reputed books of 2500+ titles on different subjects including Science, Engineering and humanities.
- It also contains unique reference books (in the field of Mechatronics, automation and nanotechnology), not available even in institutes of national importance.
- 500+ E-books (McGraw hill, Science Direct) are available with perpetual licences and 1000+ E-books (Springer, Wiley, and Pearson etc.) are under the process of purchasing.
- Online journal subscription with 2000+ journal access of springer, ASME, Elsevier Science Direct and Nature journal from Springer Nature through e-SodhSindhu.
- Member and access of National Digital Library, MHRD.
- IEEE subscription in process
- Reprographic Facility.
- Digital library equipped with 16 multimedia Pcs.



➤ **SENSORS, DRIVES AND CONTROLLAB**

Equipped with different type of sensors, hydraulic drives & pneumatic drives and actuators.

➤ **DESIGN AND SIMULATION LAB**

Equipped with high end specification computer systems loaded with ANSYS, Solid Works, Matlab softwares with research licences





➤ **Industrial Automation Lab (under Installation)**

This lab covers the industrial automation and robotics application in industry. It has different industrial machines including CNC turning, CNC Milling, Surface grinder, Automatic storage and retrieval system (ASRS), Automated Guided Vehicle (AGV), automatic loading/unloading gantry and robotic arm, SCARA robot, vision inspection system etc. This lab is helpful in understanding the application of robots and controllers in industrial automation.



➤ **Robotic Center (under Installation)**

A Robotic Center contains three KUKA (a Germany Company) robotic cells namely (i) Material handling cell (helpful in understanding the picking and placing of the jobs from one place to another), (ii) Welding cell (helpful in understanding the use of robot in welding applications) (iii) bare robot cell which may be used by the students for different applications.



➤ **3D Printing Lab (under Installation)**

The 3D printing lab contains two different types of printers. These printers are Fused Deposition Modeling (FDM) printer and Multijet printer. These printers are capable of converting the polymers into different three dimensional shapes.





➤ **NANOTECHNOLOGY RESEARCH LABS**

The Nanotechnology program's current curriculum and research topics include the fields of nanotechnology, not limited to nanoelectronics, MEMS/NEMS, nanomaterials and devices, photonics, solar cells/ batteries/computational nano-engineering. State-of-the art labs and facilities are in place or being established taking account the whole spectrum of necessary processes which complete the whole cycle of NANO – Science to Technology Cycle.

1. Material Chemistry and Synthesis Laboratory

The main goal of Materials Chemistry and Synthesis Laboratory is to experimentally explore the novel materials at the interface of Physics, Chemistry and Biology. It has instruments, equipments and systems which can transform or help to transform the materials into desired SCALE/DIMENSION (i.e, micro- and nano- scale)



2. Micro and Nano Characterization Facility (Under Construction)

The Micro and Nano Characterization facility (an ISO Class 7 cleanroom facility to accommodate the latest industry standards) plans to offer a wide variety of ultra-low dimensional material and device characterization services enabled by the presence of highest grade and quality of characterization equipment. It has following instruments and many more,

- FESEM , AFM, STM, TEM, XRD, UV-Vis-NIR and FTIR etc. spectrophotometers, Ellipsometry, Raman spectroscopy, photoluminescence spectroscopy, Thermogravimeter etc.





3. Device Testing Facility *(Under Construction)*

This facility has been designed (proposedly) with an understanding that tested/comply under certain research laboratory environment for the required parameters essential for its successful deployment at various levels and in different applications. This facility is proposed with the following instruments procurement,

- Impedance and Vector analysers, Magnetic Property Measurement System, Micro - System Analyzer, Non-contact Optical profilometer etc.



4. Battery Testing Facility *(Under Construction)*

Development of efficient materials and its application for storage of renewable energy has become the heart of energy research. Battery is one of the electrochemical energy storage devices. In battery, two different types of electrodes are assembled which convert chemical energy into electrical energy and

store as power source for future. Nanotechnology is a better option to prepare efficient electrode materials for battery. Here we are developing one dedicated laboratory for battery fabrication (coin cell, pouch cell and cylindrical cell for Li-ion/Na-ion/Li-S/Zn-air) and testing. Not only rechargeable battery, we are developing laboratory facility for supercapacitor and hybrid battery also. In future, researchers and students can develop materials, fabricate cell and test the performance of cell at our own laboratory.





Activities @ CAS

1. Memorandum of Understanding (MoU)

- a) The Nanotechnology program at CAS signed MoU with CeNSE, IISc., Bangalore for student exchange program and the use of research facilities to carry out collaborative research.
- b) MoU with Entuple Technologies Bangalore for summer internships, students training and conduct of seminars, guest lectures.
- c) Letter of Intent between Estonian University & CAS, AKTU for providing opportunities to students regarding their higher studies, specially in the field of Cyber Security.



2. CAS organised state level Dr. APJ Abdul Kalam Tehnical, Literary & Management Fest on 15 - 16 Feb. 2019.

In this event the winners of 8 Zones of U.P. have participated in 11 types of events including robowar, roborace, quiz etc.





3. Workshops/FDPs/Seminars/Lectures

A. Five Days Faculty Development Program jointly organized by AKTU and AICTE on “Emerging Trends in Networking (ETN-2019)” from 24th - 28th June 2019

The program was aimed to provide a platform for expert technical exchange regarding the advance technologies and innovations in the field of communication networks. The program has drawn attention of faculty towards the field of networking. The mandate of the FDP was to create awareness among the faculties associated to AKTU in the field of emerging networking technologies and to motivate them to contribute in high-end research activities in the field. The focus of the course was on principles, architectures, and protocols used in modern networked systems.



B. Cyber City Training Program

Advance Ethical Hacking of Cyber City & IoT Devices from 7th -11th June 2019

This training program was the second training program for the training of the students over the cyber city simulator. The program is attended by the students of Centre for Advanced Studies AKTU and the instructors from the industry has covered the following topics like, Introduction – Ethical Hacking, Ethical Hacking Methodology, Hacking Networks & Systems, Web Application Security, Denial of Services, Social Engineering, IoT Basics, A few case studies, IoT Data Integrity etc.





C. A Short Term Training course for Faculty Development jointly organized by AKTU and AICTE on Manufacturing Technology & Automation(MTA-2019) from 27th to 31st May 2019.

The program was conducted with an aim to provide a platform for expert technical exchange regarding the advance technologies and innovations in the field of Manufacturing Technology & Automation. The program would draw attention of faculty members towards the field of manufacturing technology. The program looks for significant contributions to the manufacturing sciences, engineering and automation in theoretical and practical aspects. The FDP aims to equip the participants with every skill needed to pursue sophisticated research in Manufacturing Technology and Automation field, in pursuit of a bright future of India in the international Manufacturing and Automation arena.



D. Faculty Development Program organized from 2nd to 6th April 2019 on Emerging Technologies Trends in Micro & Nano-electronics.

Some of the Speakers were: Prof. S. A. Shivashankar, Dr. Ranjit Sai, Dr. Sanjeev Srivastava, Prof. Rudra Pratap, Dr. Anshu Gaur, Dr. Prabhat Dwivedi, Prof. Navakanta Bhatt, Prof. Debabrata Goswami, Dr. Manish Hooda, Dr. Girish Kunte etc.





FACULTY ACHIEVEMENTS

1. Prof. Manish Gaur

- a. Priyanka Gupta and Manish Gaur, "A Routing Calculus with Distance Vector Routing Updates" International Journal of Advanced Computer Science and Applications (IJACSA).
- b. Priyanka Gupta and Manish Gaur, "A Comprehensive Study of Process Calculi with Routing Tables" International Journal of Advanced Computer Research (IJACR).
- c. Parul Yadav, Manish Gaur, Modelling of Support Vector Machine for Intrusion Detection System in Ad-hoc Networks using R Programming, Accepted for publication in Proc. of World Conference on Smart Trends in Systems, Security and Sustainability (WS4 2019), London, UK, 30th - 31st July, 2019.
- d. Parul Yadav, Manish Gaur, A Process Calculi for Intrusion Detection System in Mobile Ad-hoc Networks, Journal of Communications, vol. 13, no.11, pp. 635-647, November 2018.
- e. Annie Irfan, Manish Gaur, Surya Prakash Tripathi. A Study of Process Calculus for Formal Verification and Analysis of Security Protocol: Proceedings in 4th International Conference on Applied and Theoretical Computing and Communication Technology, Sep 2018. IEEE.

2. Prof M K Dutta

Funded Proposals:

- a. Title of the Proposal: Intelligent Stethoscope: A low cost device based on body auscultation for Early Medical diagnosis of Heart, Lung and Prenatal health. Principal Investigator: M.K. Dutta, Funding Agency: TDT Division, Biomedical Device and Technology Development (BDTD), Department of Science and Technology, Gov. of India. Duration: 2 years. Value: 40 Lakhs
- b. Title of the Proposal: Assistive device to impart perceptual ability to visually-impaired using intelligent scene captioning. Principal Investigator: M.K. Dutta. Funding Agency: SEED Division, Technology Interventions for Disabled and Elderly Programme, Department of Science and Technology, Gov. of India. Duration: 3 years. Value: 39.25 Lakhs

Collaborative Research Papers:

The following collaborative research papers with scientists from Spain is accepted in International Conference on Applications of Intelligent Systems", APPIS 2019, held on 7-12 January 2019 in Las Palmas de Gran Canaria, Spain, European Union. APPIS 2019 is organized by the University of Groningen and University of Las Palmas de Gran Canarias. The papers of this conference will be published in ACM Digital Library.

- A paper co-authored by Astha Singh and Shubhi Srivastava (M.Tech CSE – 2018-20 Batch) titled "Automatic Framework for Extraction of Red Lesion Using Gabor Filter from Fundus Image"
- A paper co-authored by Mansi Bhatnagar (M.Tech CSE – 2018-20 Batch) titled – "Machine Learning Based Improved Automatic Diagnosis of Cardiac Disorder"



- A paper co-authored by Vedant Singh (M.Tech CSE – 2018-20 Batch) titled – “Watermarking Scheme for Unique Patient Identification Tags in Fundus Images for Tele-Ophthalmologic Applications”

Students Papers:

Following 6 papers co-authored by Students of Centre for Advanced Studies in collaboration and co-authored by European Scientists are accepted in 42nd International conference on Telecommunication and Signal Processing (TSP-2019) held in Hungary in July 2019. TSP 2019 is organized by eighteen universities from Czech Republic, Hungary, Turkey, Croatia, Taiwan, Japan, Slovak Republic, Spain, Bulgaria, France, Romania, Slovenia, Greece, and Poland. All these papers will be included in IEEE Xplore Digital Library and will be indexed in Scopus and Web of Science.

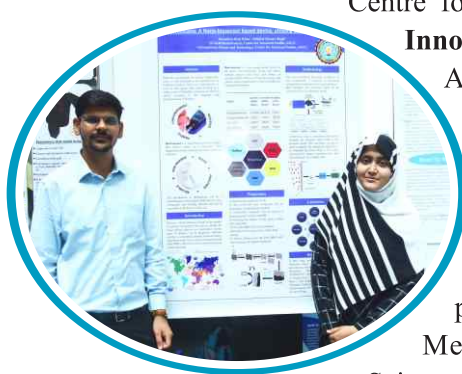
1. Shubhi Srivastava, Astha Singh, Anjali Yadav, Malay Kishore Dutta, Kamil Riha, Jan Dorazil, “Automatic Extraction of Micro-aneurysms and Haemorrhages from Digital Fundus Image”.
2. Arpit Gupta, Arshdeep Kaur, Malay Kishore Dutta & Jiri Schimmel, “ Article Title: Perceptually Transparent & Robust Audio Watermarking Algorithm Using Multi Resolution Decomposition & Cordic QR Decomposition”
3. Ankita Dwivedi, Ankit Kumar, Malay Kishore Dutta, Radim Burget, Vojtech Myska, “An Efficient and Robust Zero-Bit Watermarking Technique for Biometric Image Protection”.
4. Areeba Irshad, Ritesh Maurya, Malay Kishore Dutta, Radim Burget, Vaclav Uher, “Feature Optimization for Run Time Analysis of Malware in Windows Operating System using Machine Learning Approach”.
5. Anam Fatima, Ritesh Maurya, Malay Kishore Dutta, Radim Burget, Jan Masek, “Android Malware Detection Using Genetic Algorithm based Optimized Feature Selection and Machine Learning”.
6. Ritesh Maurya, Malay Kishore Dutta, Kamil Riha, Petr Kriz, “An Image Processing based Identification of Fish Exposed to Polluted Water.

3. Dr. Anuj K Sharma

- i. Received a research grant of Rs. 5.0 Lakh as Principal Investigator (Dr. Rabesh Kumar Singh as Co-PI) from AKTU Lucknow under Visvesvaraya Research Promotion Scheme.
- ii. Received a research Grant of Rs. 3.0 Lakh as Principal Investigator (Dr. Rabesh Kumar Singh as Co-PI) under Collaborative Research and Innovation Program (CRIP) of AKTU TEQIP-3 scheme.
- iii. Sharma et al. “Tribological investigation of r-GO additived biodegradable cashew nut shells liquid as an alternative industry lubricant” Tribology International. (SCI publication)
- iv. Sharma et al. “Physiochemical Properties of Hexagonal Boron Nitride Blended Coconut Oil” Available at SSRN 3321035.
- v. Contributed a chapter to a book entitled “Nanofluids and their engineering applications” which will be published under the imprint of CRC Press/Taylor & Francis Group, USA. *Chapter Title:* Hybrid nanoparticles enriched cutting fluids in machining processes. Co-authors: Dr. Amit Rai Dixit, Dr. Arun Kumar Tiwari, Dr. Rabesh Kumar Singh.



Innovative Idea Contest at Centre for Advanced Studies, AKTU



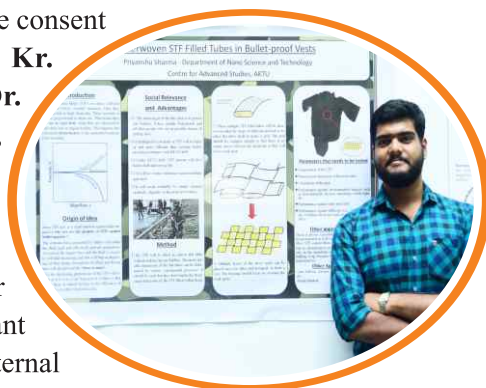
Centre for Advanced Studies, AKTU has organised an **Innovative Idea Contest, 2019 (Phase I)** on 27th April, 2019 for the MTech Students. Objective of the program was to encourage the M.Tech. Students from CAS towards innovation and Technology. This could be a unique opportunity for the students to explore their concepts and ideas. The centre is planning to organize this program twice a year. Students from Mechatronics, Nanotechnology and Computer

Science and Engineering have actively participated in the event. **Prof. Manish Gaur, Director, CAS** has inaugurated the program and constantly support to conduct this event. The idea of conducting this contest was proposed by **Prof. M. K. Dutta, Dean R&D, AKTU** and with the consent

of Hon'ble Vice chancellor **Prof. Vinay Kr. Pathak**, the dream turned into reality. **Dr. Ramkrishna Sahoo, DST INSPIRE Faculty,**

Nanotechnology has coordinated this whole event with the help of **Dr. Piyush Jaiswal, Coordinator, Nanotechnology,** **Dr. Anuj Sharma, Coordinator, Mechatronics** and other faculty members. **Dr. Atul Thakur, Assistant Professor, IIT Patna** was present as an external judge of this event. Out of the 10 interesting ideas, **Mr.**

Abhijit Kumar Singh (Nanotechnology, MTech 1st year) and **Ms. Juwairiya Siraj Khan** (Mechatronics, MTech, 1st Year) got the 1st prize on their idea on BIOTRONICANO. **Mr.**



Priyanshu Sharma, (Nanotechnology, MTech 1st year) got the 2nd prize for his idea on the making of bullet-proof vests using Shear Thickening Fluids (STF).

Mr. Abhishek Kaushal, (Mechatronics, MTech, 1st Year) got the 3rd prize for his idea on artificial intelligence for blind. Institute has announced prize money for the winners (1st Prize - Rs. 10,000, 2nd Prize - Rs. 7,500 and 3rd Prize - Rs. 5,000) and funding up to Rs. 50,000 per abstract for prototype preparation and patent submission.

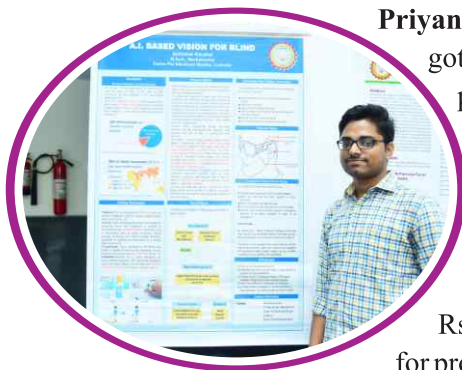




Photo Gallery

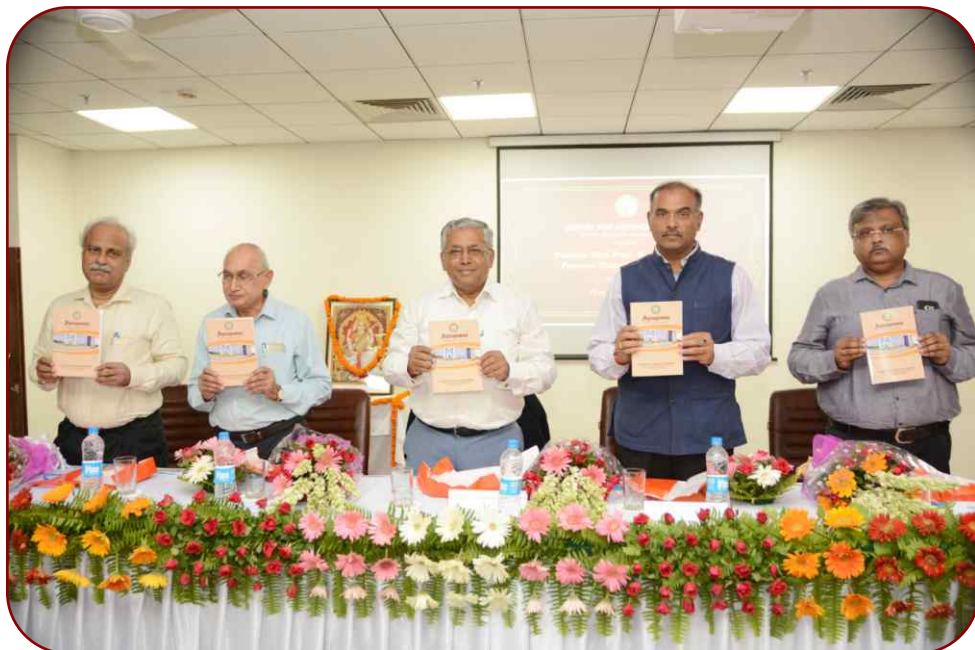
Inauguration of Google Code Lab by Hon'ble Minister of Technical Education, U.P. Govt. Shri Ashutosh Tandon "Gopal Ji" in presence of AKTU Vice Chancellor, Prof. Vinay Kumar Pathak & Prof. Manish Gaur, Director, CAS.





Visits of eminent personalities@CAS





Release of Agragaman 2018 (Annual Progress Report of CAS) during the celebration of First Raising Day of Centre for Advanced Studies



Vice Chancellor AKTU, Prof. Vinay Kumar Pathak addressing the participants of State Level Tech Fest on 15 Feb. 2019

Student visits to Industry/Research Labs



Expert Talks



Innovative Idea Contest @ CAS





**Dr. APJ Abdul Kalam Technical, Literary & Management
State Level Fest - 2019 @CAS on 15 - 16 Feb. 2019**





Board of Studies meeting of M.Tech. Manufacturing Technology & Automation and Computer Science & Engineering Programs.



Other Activities



“Small Aim is a
Crime
Have a
*Great
Aim*”



Centre for Advanced Studies

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